

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular End Semester Examination – Summer 2023**

**Course: B. Tech.**

**Branch: Civil Engineering**

**Semester: VI<sup>th</sup>**

**Subject Code & Name: BTCVOE605H Applications of Remote Sensing & GIS**

**Max Marks: 60**

**Date: 21/07/2023**

**Duration: 3.00 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Draw a suitable diagram wherever necessary.

Marks

**Q. 1 Solve Any Two of the following.**

- A) Write short notes on: - **6**
- a. Rayleigh scattering
  - b. Atmospheric window
  - c. Use of low frequency waves
- B) Write a detailed note on the procedure of the remote sensing technique. **6**
- C) Explain the electromagnetic spectrum and its role in remote sensing. **6**

**Q.2 Solve Any Two of the following.**

- A) Write a short note on: - **6**
- a. Swath
  - b. Whisk-broom scanning system
  - c. NAVIC system
- B) Write a difference between active and passive remote sensing. **6**
- C) Enumerate a microwave radar sensing system with its limitations and advantages. **6**

**Q. 3 Solve Any Two of the following.**

- A) Write a short note on: - **6**
- a. N. D. V.I.
  - b. Analogue image
  - c. Radiometric correction
- B) Write a detailed note on spectral signature characteristic features of vegetation and influencing factors. **6**
- C) Explain in detail the supervised classification of imageries with the procedure and its disadvantages. **6**

**Q.4 Solve Any Two of the following.**

- A) Write a short note on: - **6**
- a. Triangular Irregular Network (TIN)
  - b. D.E.M.
  - c. Attribute data
- B) Enumerate the difference between raster data and vector data. **6**
- C) Write a detailed note on Geographic Information System with its advantages and limitations. **6**

**Q. 5 Solve Any Two of the following.**

- A) Write a short note on: - **6**
- a. Universal Transverse Mercator
  - b. Geoid
  - c. Longitude
- B) Write a detailed note on Geographic Coordination System with its strengths and weaknesses. **6**
- C) Explain in detail about various techniques used for approximation of Earth. **6**

**\*\*\* End \*\*\***